# Multimodal Visual Understanding



## 1. Course Content

Run the example program to allow the robot to observe its environment and execute tasks based on instructions via text interaction on the terminal.

> [!NOTE]
>
> - The only difference between the text and voice versions here is the method of instruction input, and the text version does not include speech recognition and speech synthesis playback.

## 2. Preparation

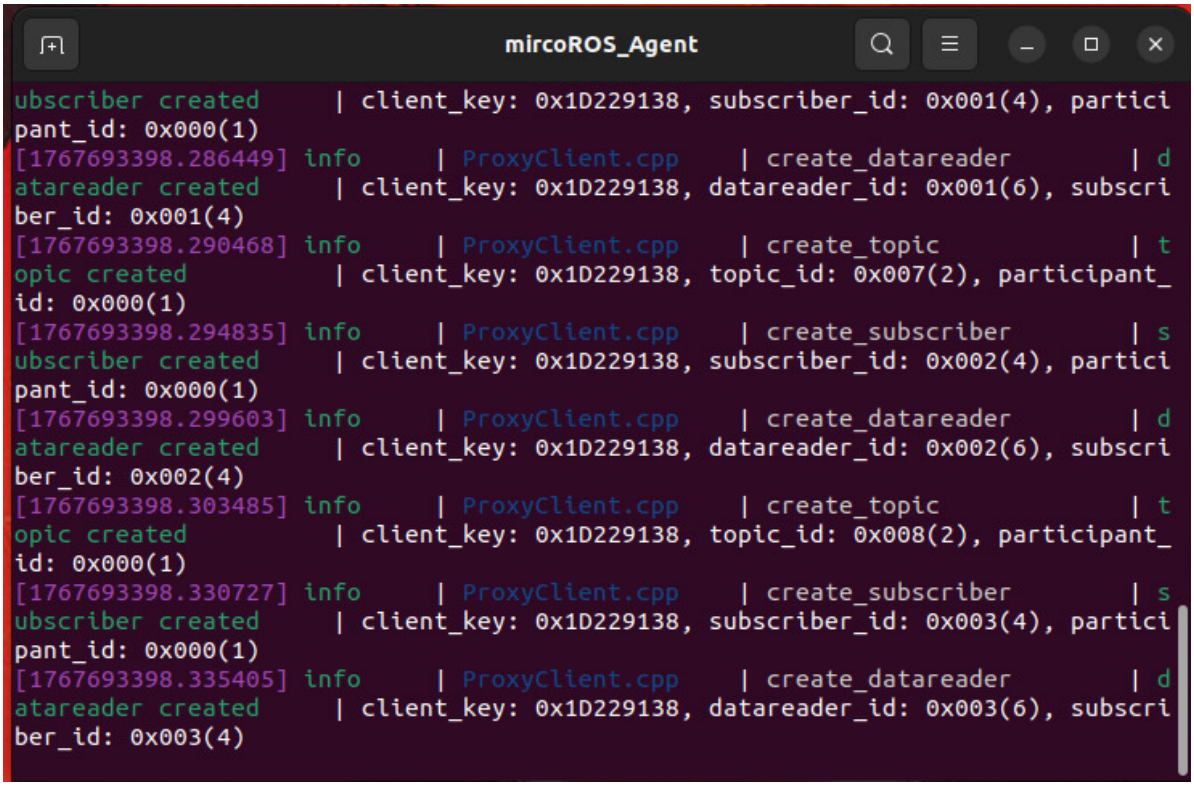
### 2.1 Starting the Agent

**Note: The agent must be started before testing any case. If it is already started, it does not need to be started again.**

Enter the following command on the vehicle terminal:

```
start_agent
```

The terminal prints the following information, indicating a successful connection:




# 3. Running the Case

## 3.1 Starting the Program

**For Raspberry Pi 5 users, if the ROS container is not running, you need to start it first.**

```
bringup_m3
```

**Open the terminal inside the container.**

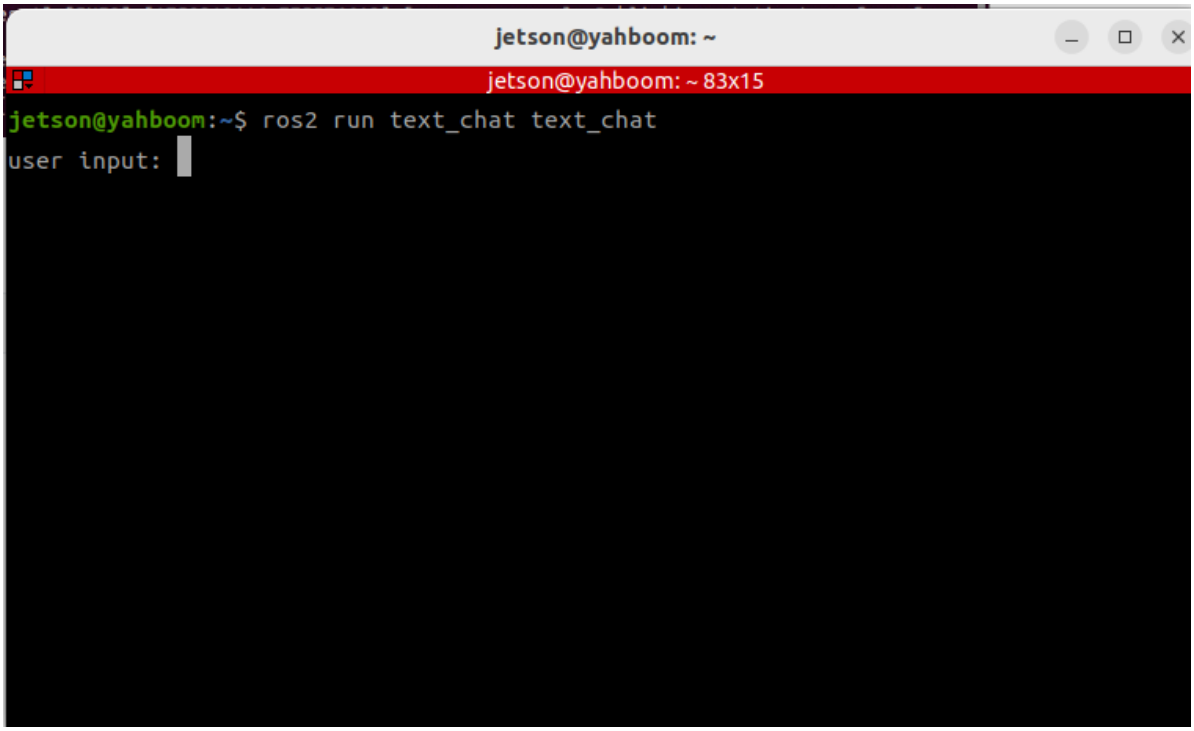
```
exec_m3
```

- Open the terminal on the vehicle terminal and enter the following command:

```
ros2 launch m3_bringup llm_agent_control.launch.py text_chat_mode:=True
```

- Start the text interaction node:

```
ros2 run m3_common_demo text_chat
```



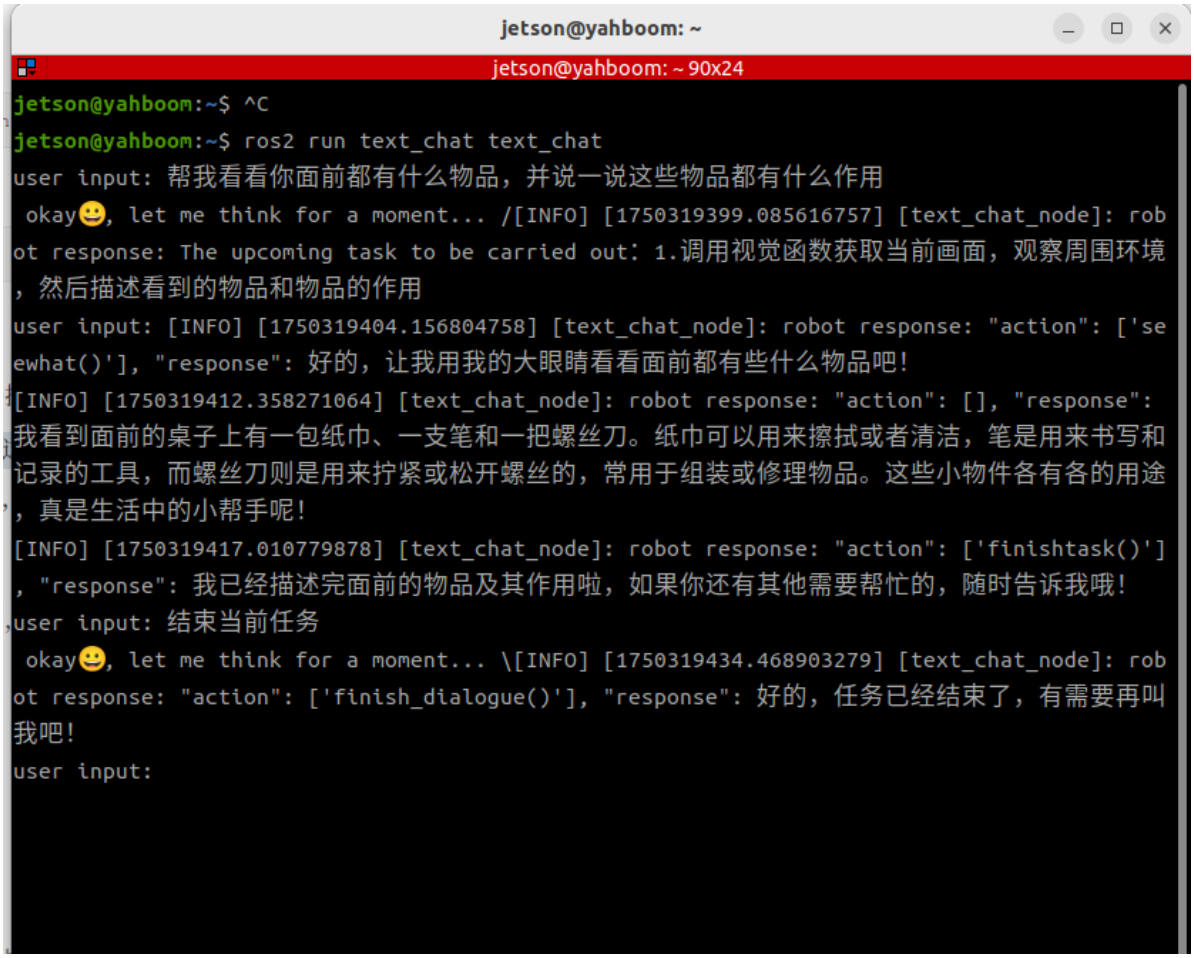

## 3.2 Test Cases

Two reference test cases are provided here. Users can create their own test commands.

- Please look at the items in front of you and tell me what they are and what their functions are.

### 3.2.1 Case 1

Enter the test case in the text interaction terminal. After the model thinks, it replies to the user and executes actions according to the user's instructions.




After completing its task, the robot will enter a waiting state. During this time, you can continue the conversation. Commands given at this point will be directly passed to the execution layer's large model, and all conversation history will be permanently retained. You can enter the command "End current task" again to end the current task cycle and begin a new one.

# 4. Source Code Analysis

- Robot Action Source Code Path:

```
~/yahboomM3_ws/code_ws/src/multi_brains/multi_brains/action_service.py
```

- Model Service Source Code:

```
~/yahboomM3_ws/code_ws/src/multi_brains/multi_brains/model_service.py
```

- The main program implementing the robot's visual observation function is the `seewhat` method in the `action_service.py` program:
- The function implementation is: save and display an image from the latest perspective
- Then send a request to the `model_service` node of the model service, requesting the multi_brains agent in the dify to provide feedback on the image.

```
    def seewhat(self):
        """
        Save the current view image and send it back to the dify-agent.
        """
        self.save_single_image()
        msg=LlmRequest()
        msg.llm_request=self.actionlog.get_text("image_feedback")
```

```
        msg.robot_feedback=True
        self.llm_request_pub.publish(msg)
        return None

    def save_single_image(self):
        """保存一张图片 / Save a single image"""
        cv_image = self.bridge.imgmsg_to_cv2(self.image_msg, "bgr8")
        cv2.imwrite(self.image_cache_path, cv_image)
        time.sleep(0.05)
        display_thread = threading.Thread(target=self.__display_saved_image)
        display_thread.start()


    def __display_saved_image(self):
        """
        显示已保存的图片4秒后关闭窗口 / Display the saved image for 4 seconds before
closing the window
        """
        try:
            img = cv2.imread(self.image_cache_path)
            if img is not None:
                cv2.imshow("Saved Image", img)
                cv2.waitKey(4000)  # 等待4秒 / Wait for 4 seconds
                cv2.destroyAllWindows()
            else:
                self.get_logger().error(
                    "Failed to load saved image for display."
                )  # Loading saved image for display failed...
        except Exception as e:
            self.get_logger().error(f"Error displaying image: {e}")  # Error
displaying image...
```

- Additionally, in model_service.py, the `llm_request_callback` callback function is used to receive requests to access the `multi_brains` agent.
- If the `llm_request` field in the request information is an image request, then a list `[msg.llm_request,'image_request',True]` is constructed and added to the model request processing queue.

```
    def llm_request_callback(self, msg:LlmRequest):
        '''
        话题回调函数,接收调用模型请求并放入队列中 / Topic callback function, receive
model request and put into queue
        '''
        if self.debug_mode: self.get_logger().info(f"robot_feedback:
{msg.robot_feedback},llm_request:{msg.llm_request}")
        if msg.robot_feedback:
           # Robot Feedback Request
            if msg.llm_request ==self.syslog.get_text("image_feedback"):

 self.llm_handler_queue.put([msg.llm_request,'image_request',True])
            elif msg.llm_request =="finish":
                # Upon receiving the finish command from dify-agent, the current
task cycle ends, which clears the historical context and begins a new task
cycle.
                self.clear_request_queue()  # Clear the request queue
```

```
                self.dify_llmclient.reset_conversation()  # Reset Session
            else:
                #Standard robot feedback results
                if self.debug_mode:
self.get_logger().info(self.syslog.get_text("system_log_4"))

 self.llm_handler_queue.put([msg.llm_request,'text_request',True])

        else:# Model requests from other sources
            self.llm_handler_queue.put([msg.llm_request,'text_request',None])
```

- In the `handle_llm_thread` method of `model_service.py`, the response mode is distinguished by the `self.text_chat_mode` variable. If it is only text interaction mode, only text response will be given.

```
    def handle_llm_thread(self)->None:
        '''
        处理模型请求/ Handle model request
        '''
        while True:
            if not self.llm_handler_queue.empty():#The queue is not empty,
process the model request.
                request_query = self.llm_handler_queue.get()
                if self.debug_mode: self.get_logger().info(f"Processing LLM
request: {request_query}")

                if request_query[1]=='text_request':
                    '''text request'''

 result=self.dify_llmclient.chat(request_query[0],robot_feedback=request_query[2
])
                elif request_query[1]=='image_request':
                    '''vision + text request'''

 result=self.dify_llmclient.chat(request_query[0],image_path=self.image_cache_pa
th,robot_feedback=request_query[2])

                if result[0]:
                    if  not self.text_chat_mode and self.asr_detect.record_flag
: continue
                    split_result=self.extract_actions(result[1])
                    if split_result is None: continue

 action_list,llm_response,decision_plan=self.extract_actions(result[1])
                    if decision_plan is not None:

 self.get_logger().info(Fore.YELLOW+self.syslog.get_text("system_log_3",decision
_plan=decision_plan)+Fore.RESET)
                    self.get_logger().info(Fore.YELLOW+f'"action":
{action_list},"response": {llm_response}'+Fore.RESET)

                    if  not self.text_chat_mode:#Voice reply
                        if
self.tts_engine.synthesize(llm_response,self.tts_out_path) :
                            self.play_audio(self.tts_out_path)
```

```
                        else:
                            self.get_logger().error(Fore.RED+"Speech synthesis
failed. Check whether the TTS model is available"+Fore.RESET)
                    else:#Text Reply
                        if decision_plan is not None:

 self.text_pub.publish(String(data=self.syslog.get_text("system_log_3",decision_
plan=decision_plan)))
                        self.text_pub.publish(String(data=f'"action":
{action_list}, "response": {llm_response}'))

                    if action_list!=[]: self.send_action_service(action_list,
llm_response)
                else:
                    self.get_logger().error(Fore.RED+f"The model request failed.
Check whether the dify or AI model is normal.\
                                            Error Log:{result[1]}"+Fore.RESET)
            else:
                time.sleep(1.0)# Sleep for 1 second if there is no request
```

- During model engine initialization, speech recognition and speech synthesis models are only loaded in voice interaction mode.

```
    def init_largemodel(self):
       #Initialize Dify

 self.dify_llmclient=Dify_LLM_Client(api_key=self.dify_api_key,base_url=self.dif
y_base_url)
        result=self.dify_llmclient.test_connection()
        if result[0]:
            self.get_logger().info("Dify LLM connection successful")
        else:
            self.get_logger().error(Fore.RED+f"Dify LLM connection failed,\
                                    Please check whether DIY dify has been
started, API and BASE_URL settings\
                                    Error log:{result[1]}"+Fore.RESET)
        llm_handler_thread =
threading.Thread(target=self.handle_llm_thread)#Start the dify thread to handle
business related to large model requests.
        llm_handler_thread.daemon = True
        llm_handler_thread.start()

        if not self.text_chat_mode:  # Voice Chat Mode
            #Initialize ASR
            self.asr_detect=ASR_Detect(self.llm_handler_queue,self.config_file,
pygame_lock=self.pygame_lock)
            if not self.asr_detect.init_ASR_Detect() :
                self.get_logger().error(Fore.RED+"Failed to initialize
ASR_Engine"+Fore.RESET)

            asr_thread =
threading.Thread(target=self.asr_detect.asr_detect_run)#Start the ASR thread to
handle wake-up and speech recognition related tasks.
            asr_thread.daemon = True
            asr_thread.start()
```

```
            #Initialize TTS

 self.tts_out_path=os.path.join(os.path.expanduser('~'),"yahboomM3_ws","user_ws"
,"cache", "tts_output.wav")   # 语音合成输出路径 / TTS output path

            if self.config_param.get("USE_OLINE_TTS", False):
                tts_supplier=self.config_param.get("TTS_SUPPLIER")

                if tts_supplier=="aliyun":
                    self.tts_engine=TongyiTTS(self.config_file)
                    if self.tts_engine.init_tts_engine() :
                        self.get_logger().info(Fore.BLUE+"Tongyi TTS_Engine
initialized successfully"+Fore.RESET)
                    else:
                        self.get_logger().error(Fore.RED+"Tongyi TTS_Engine
initialized failed"+Fore.RESET)
                elif tts_supplier=="baidu":
                    self.tts_engine=BaiduTTS(self.config_file)
                    if self.tts_engine.init_tts_engine() :
                        self.get_logger().info(Fore.BLUE+"Baidu TTS_Engine
initialized successfully"+Fore.RESET)
                    else:
                        self.get_logger().error(Fore.RED+"Baidu TTS_Engine
initialized failed "+Fore.RESET)

                elif tts_supplier=="xunfei":

 self.tts_out_path=os.path.join(os.path.expanduser('~'),"yahboomM3_ws","user_ws"
,"cache", "xunfei_tts_output.mp3")

 key_path=os.path.join(os.path.expanduser('~'),self.config_param.get("XUNFEI_KEY
",""))

 xunfei_encrypted_key=os.path.join(os.path.expanduser('~'),self.config_param.get
("XUNFEI_ENCRYPTED",""))

 self.tts_engine=XunfeiTTS(key_path,xunfei_encrypted_key,self.config_file)

                    if self.tts_engine.init_tts_engine():
                        self.get_logger().info(Fore.BLUE+"Xunfei TTS_Engine
initialized successfully"+Fore.RESET)
                    else:
                        self.get_logger().error(Fore.RED+"Xunfei TTS_Engine
initialized failed"+Fore.RESET)

            else:
                self.tts_engine=PiperTTS(self.language)
                if self.tts_engine.init_tts_engine() is not False:
                    self.get_logger().info(Fore.BLUE+"Piper TTS_Engine
initialized successfully"+Fore.RESET)
                else:
                    self.get_logger().error(Fore.RED+"Piper TTS_Engine
initialized failed"+Fore.RESET)
```